

Bundled Feature Encoding: A Shift-Tolerant Image Representation for Hyperdimensional Computing

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Abstract

Hyperdimensional Computing (HDC) is a brain-inspired, lightweight machine learning paradigm that represents information using high-dimensional vectors. While HDC has shown significant promise in computer vision, the field currently lacks a robust and consistent framework for encoding image data. Existing whole-picture pixel-encoding methods are highly sensitive to spatial shifts, viewpoint variations, and background redundancy, strictly limiting their viability for real-world robotic perception. To address these limitations, this paper introduces Bundled Feature Encoding (BFE), a novel framework comprising three distinct methods: Column-Based, Row-Bundled, and Block-Bundled Feature Encoding. By binding distinct pixel segments into robust spatial features before bundling them into a single global image hypervector, BFE captures structural information that remains stable under perspective shifts and inherent camera noise. Experimental evaluations using real-world robotic navigation data demonstrate that one-dimensional BFE approaches, particularly row and column bundling, significantly improve similarity preservation during continuous movement while reliably maintaining orthogonality for dissimilar scenes. These results highlight BFE’s potential as an efficient, shift-tolerant encoding paradigm for autonomous mobile robot vision and other resource-constrained perception systems.

Keywords: Hyperdimensional computing, Vector Symbolic Architecture, Computer Vision, Image Encoding, Feature Recognition

1 Introduction

Hyperdimensional Computing (HDC), also known as Vector Symbolic Architecture (VSA), is an emerging computational paradigm inspired by the distributed representations found in biological neural systems [Kleyko et al., 2023a, Plate, 1995, Kanerva, 1996, Gayler, 1998]. Information is encoded using very high-dimensional vectors whose statistical properties enable robustness to noise, efficient similarity comparison, and lightweight learning. These characteristics make HDC an attractive alternative to conventional machine-learning approaches, particularly for resource-constrained platforms such as mobile robots and embedded systems.

Despite growing interest in applying HDC to vision tasks, image encoding remains one of the least mature areas in the field [Kleyko et al., 2023b]. Existing methods typically fall into three categories: pixel-based encoding, feature-enhanced encoding, and hybrid neural-HDC approaches. Pixel-position encoding, while conceptually simple, suffers from extreme sensitivity to spatial shifts, viewpoint changes, and background redundancy. Hybrid methods improve accuracy but sacrifice the computational efficiency that makes HDC appealing. Feature enhanced encoding methods have mainly been used to develop inputs for image classifiers which calculate the most likely match between hypervectors. Hybrid methods have been developed as high efficiency alternatives to deep neural networks and perform moderate to well with far less computational cost. Although all these HDC image encoding strategies often use feature extraction, they may not work well for the direct image matching required for tasks such as Visual Place Recognition (VPR), a process where a robot determines its current location by matching its live camera view against a stored database of previously seen

images. As a result, there is a lack of practical frameworks for hyperdimensional encoding of images in a way that preserves meaningful structure while remaining lightweight for robotic applications. Therefore, this paper introduces Bundled Feature Encoding (BFE), a novel technique designed to address these limitations by capturing stable features that remain consistent across viewpoint variations. The method aims to improve robustness while maintaining the efficiency required for real-time robotic perception.

2 Hyperdimensional Computing

Binary hypervectors are a predominant representation scheme in Hyperdimensional Computing (HDC), relying on four core operations: similarity measurement, bundling, binding and permutation [Gayler, 1998]. Binary hypervector representations are generally preferred over bipolar, real, and complex hypervectors because computing devices can execute binary operations much more efficiently. An example of a binary vector is illustrated in Equation (1), when it’s dimension d (number of bits) is sufficiently large (typically 10,000), it is called a hypervector.

$$\mathbf{v} \in \{0, 1\}^d : [\mathbf{v} = (0, 0, 1, 0, 1, 1, 0, \dots, 0)] \quad (1)$$

Hypervectors possess unique mathematical properties that enable robust, highly distributed information encoding. For binary hypervectors, comparing vectors is highly hardware-efficient using normalised Hamming similarity, which measures the proportion of matching bits. In high-dimensional spaces, randomly generated vectors are inherently orthogonal; they are completely uncorrelated and typically share 50% of their bits (a similarity of 0.5). A similarity of 1 indicates identical vectors, whereas a similarity of 0 means the vectors are exact complements (a bitwise XOR of the two would yield a vector of 1s).

Bundling creates a single hypervector that represents a set or collection of multiple hypervectors. For binary hypervectors, bundling is achieved via component-wise majority voting. In the event of a tie, the result is assigned either stochastically (randomly) or deterministically (e.g., biased toward zero). This process ensures the resulting hypervector remains the same dimensionality as its members. In a bundled hypervector, each component (bit for binary) matches the corresponding component of its constituents with a probability greater than 0.5. Consequently, the bundled representation maintains a measurable similarity to each of its constituent vectors. Although individual bits may be flipped by the majority-vote process, acting as local interference, the massive scale of high-dimensional space ensures that the overall similarity remains statistically stable and reliable. This “holographic” property allows bundling to reliably encode relationships and sets.

Binding is the operation used to associate two hypervectors, creating a compositional representation of a relationship. This is typically used to form role/filler pairs, key/value associations, or symbol/attribute links, such as binding the attribute “red” to the object “circle”. For binary hypervectors, binding is implemented as a bitwise Exclusive OR (XOR) operation as illustrated in Equation (2).

$$\mathbf{C} = \mathbf{A} \oplus \mathbf{B} \quad (2)$$

The binding operation is characterised by two essential properties: orthogonality and invertibility. First, binding acts as a similarity-destroying (information-scrambling) transformation where the resulting hypervector $\mathbf{C}$ is dissimilar to both operands $\mathbf{A}$ and $\mathbf{B}$. Specifically, if $\mathbf{B}$ is a random hypervector, $\mathbf{C}$ shares approximately 50% of its bits with $\mathbf{A}$, rendering the vectors orthogonal in high-dimensional space. This stochastic behaviour ensures that binding binary hypervectors produces a result that does not interfere with the identity of its constituents. Second, the operation is invertible because XOR is its own inverse. This makes unbinding straightforward: knowing one operand allows recovery of the other via $\mathbf{A} = \mathbf{C} \oplus \mathbf{B}$, enabling the reliable retrieval of encoded information.

Permutation in hyperdimensional computing provides a consistent, reversible method for encoding sequence and structure. While mathematically representable by complex permutation matrices, practical HDC almost exclusively employs a simple circular shift of the vector’s components for hardware efficiency [Schlegel et al., 2022] (e.g., an order-1 shift moves (a, b, c, d, e) to (b, c, d, e, a)). Crucially, the permutation

operation is invertible, distributes over component-wise operations, and acts as an information scrambler, producing a vector that is completely dissimilar to the original.

With these foundational operations established, hyperdimensional computing can be applied to encode complex, real-world data. Whole-image Pixel Position Encoding (WPPE) encodes an image by combining pixel intensities and their coordinates [Mitrokhin et al., 2019, Kleyko et al., 2019]. A set of hypervectors is generated to represent pixel intensities (256 is typical for an 8-bit image). The entire set can be orthogonal by generating random hypervectors, or the similarity between intensities can be controlled by generating only one random hypervector for the lower bound (zero) interval. The next hypervector is then produced by randomly flipping bits in the hypervector, a flip probability can be adjusted to control how similar the neighbouring hypervectors are. This is repeated across the full set. The Hamming similarity of the hypervectors representing the 256 greyscale intensity values can be visualised using heatmaps, as shown in Figure 1. The three figures illustrate the effect of different flip probabilities on the representational similarity. Figure 1(a) uses a flip probability of 0.5, resulting in random, orthogonal hypervectors for all distinct grey values (shown as a uniform dark background with a similarity of 0.5) and a sharp, 1-pixel-wide diagonal line where identical values have a similarity of 1.0. Figure 1(b) shows that reducing the flip probability to 0.015 preserves similarity between close shades of grey while leaving non-similar shades orthogonal. Finally, Figure 1(c) demonstrates that a flip probability of 0.0045 creates a wider, smoother band of similarity across the intensity range, with only the most distant pixel intensities approaching orthogonality.

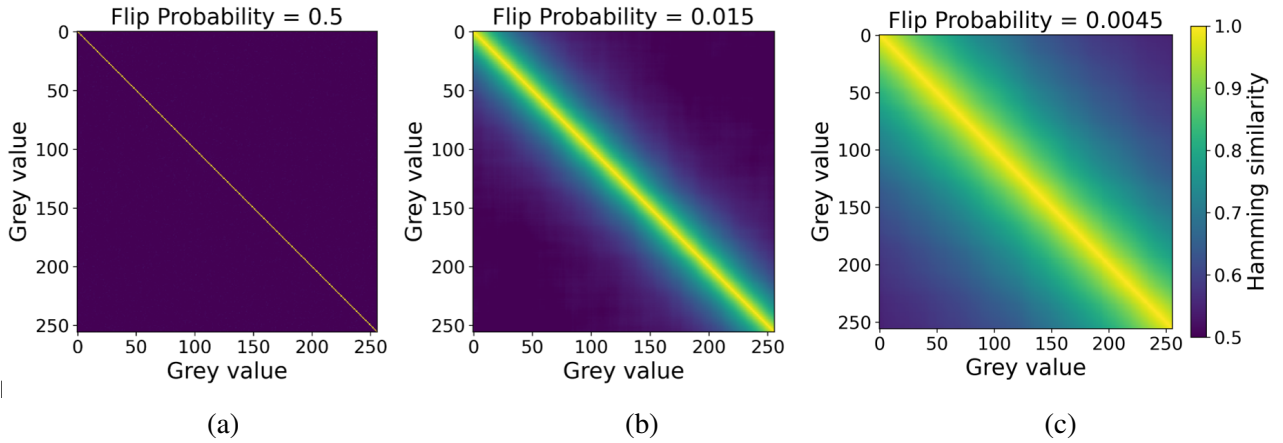


Figure 1: Heatmaps illustrating the Hamming similarity between hypervectors representing 256 greyscale intensity values at different flip probabilities.

To encode spatial positions, the base intensity hypervector for each pixel is systematically permuted (circularly shifted). The total shift amount is determined by the pixel’s x and y coordinates. To prevent spatial collisions, where different coordinate pairs might accidentally produce the same total shift, a prime number multiplier (e.g., $P = 31$) is applied to one of the axes to mathematically differentiate them. Finally, the complete image is represented as a single global hypervector by bundling all of these spatially permuted pixel vectors together using majority rule.

This WPPE approach can result in a massive number of bound features in the high-dimensional space. For larger images, the majority of the data might be considered background, leading to significant redundancy or recognition errors. The positional encoding of each pixel makes image recognition extremely sensitive to small spacial shifts of the image. Thus, the encoding fails to capture and compress meaningful semantic data. Instead, the accumulated noise dilutes the actual structural information relevant for classification and adds to the computational load. Consequently, pixel-based encoding encounters a performance ceiling, struggling significantly with classification tasks that require discernment of complex, localised features, necessitating the development of more sophisticated methods like feature enhanced and hybrid approaches. To unlock HDC’s true potential for computer vision, the encoding strategy must shift from rigidly memorising exact pixel coordinates to

binding and bundling resilient structural segments that remain stable even when the perspective changes.

3 Methodology

To overcome the extreme spatial sensitivity of traditional Whole-image Pixel Position Encoding (WPPE), this paper introduces and evaluates three novel encoding techniques based on localised structural feature detection: Column Bundled Feature Encoding (CBFE), Row Bundled Feature Encoding (RBFE), and Block Bundled Feature Encoding (BBFE). Rather than strictly encoding absolute, single-pixel coordinates, these methods bind one- or two-dimensional segments of the image using vertical columns, horizontal rows, or discrete blocks, respectively, into recognisable structural features. These extracted features are then bundled to generate a global, spatially robust image representation. The primary hypothesis is that extracting specific directional structures provides targeted invariance to viewing angle shifts; for instance, capturing only vertical features (CBFE) should yield invariance to horizontal perspective shifts, whereas horizontal features (RBFE) may prove robust against vertical panning or camera pitch changes. This study first details the mechanics of these three approaches, followed by experimental evaluation of their performance, computational efficiency, and sensitivity limits using highly similar real-world image datasets.

Across all approaches, the process begins by generating a base dictionary of 256 pixel-intensity hypervectors based on selected flip probabilities. Every pixel in the image is then mapped to its corresponding intensity hypervector. For Column-Based Feature Encoding (CBFE), we extract spatial information using a vertical sliding window. As illustrated in Figure 2, this window isolates a specific sequence of pixels, such as a 5-pixel vertical column. To preserve the spatial order within this column, a circular shift (permutation) is applied to each pixel's hypervector based on its vertical position. The top pixel's hypervector remains unchanged (zero shift), the second is shifted by one position, the third by two, and so on. These permuted hypervectors are then bound together to create a single, unique hypervector that represents that specific column feature. This sliding window moves sequentially to capture features across the entire image. Thus when processing a 30 x 30 pixel image with a 5-pixel vertical window we require 30 x 25 column features. Once all column features are extracted, their respective hypervectors are bundled together to produce a single composite hypervector that represents the entire image.

The primary advantage of this approach is its robustness to perspective shifts. By exclusively capturing vertical features, the CBFE method achieves greater invariance to changes in the viewing angle. In contrast, horizontal features tend to skew or distort when the angle changes, which disrupts similarity matching in standard pixel-based encoding methods.

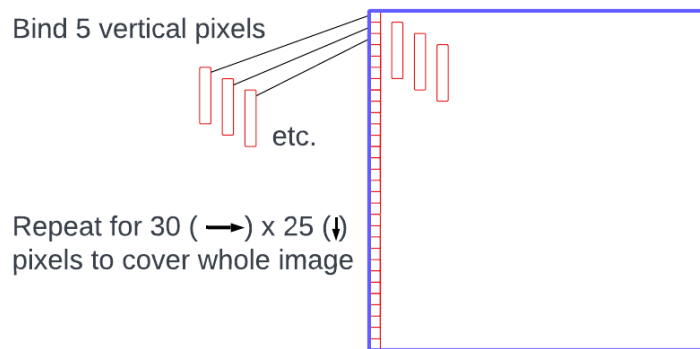


Figure 2: For CBFE, short columns of pixels are bound (5 pixel windows are illustrated in this image). This is repeated for the entire image. The full set is then bundled to produce the final image hypervector.

Row Bundled Feature Encoding (RBFE) directly adapts the principles of the CBFE technique by extracting spatial features along horizontal pixel rows rather than vertical columns. By prioritising horizontal spatial struc-

tures, RBF is specifically designed to achieve invariance against vertical movement and changes in camera pitch. Under such conditions, vertical features are prone to skewing and distortion, whereas horizontal features remain relatively stable, thereby preserving the structural integrity required for accurate similarity matching.

Block Bundled Feature Encoding (BBFE) expands upon the one-dimensional approaches of CBFE and RBF by extracting two-dimensional spatial features. This method utilises a sliding window to capture overlapping 2D blocks of pixels, such as an 8×8 pixel region, as individual features. To preserve the precise 2D spatial arrangement within each block, both the horizontal (x) and vertical (y) positions of every pixel must be explicitly encoded. Similar to Whole Picture Pixel Encoding (WPPE), this is achieved using position-dependent permutation shifts, incorporating a prime number multiplier to mathematically distinguish the x -coordinates from the y -coordinates during the shifting process. Once the pixels within a block are positionally encoded, their respective hypervectors are bound together to generate a single feature hypervector representing the structural composition of that specific 8×8 region. As the window traverses the image, the resulting feature hypervectors for all overlapping blocks are subsequently bundled to form the final composite hypervector for the entire image. This 2D block approach presents a clear theoretical trade-off in hyperdimensional system design. On one hand, it significantly enhances the representation of complex, two-dimensional structural information, theoretically making it more robust to simultaneous shifts in multiple directions. On the other hand, this high-fidelity structural capture comes at a substantially higher computational cost, as permuting, binding and bundling overlapping multi-dimensional arrays requires exponentially more operations than the 1D column or row methods.

To mitigate computational overhead during the feature extraction process, the stride of the sliding window can be adjusted. In the standard implementation, a baseline stride of 1 is utilised, meaning the window advances by a single pixel each time to capture a highly dense, overlapping set of spatial features. However, to systematically reduce the required number of hyperdimensional binding and bundling operations, larger stride intervals, for example strides of 2 and 5, can be introduced. By increasing the stride, the window skips intermediate pixels between extractions, which significantly decreases the total volume of feature hypervectors generated per image.

4 Results and Evaluation

To evaluate the encoding and recognition capabilities of CBFE, RBF, and BBFE, we utilised a subset of the Multi-Domain Dataset for Robots (MDDR) [Wozniak et al., 2025], samples of which are shown in Figure 3. This subset comprises an 800-frame video sequence captured by a robot-mounted Pi camera navigating a looping corridor. The dataset provides an excellent testbed for these encoding methods due to its subtle inter-frame movements and realistic sensor noise, such as auto-exposure adjustments and white balance drift.

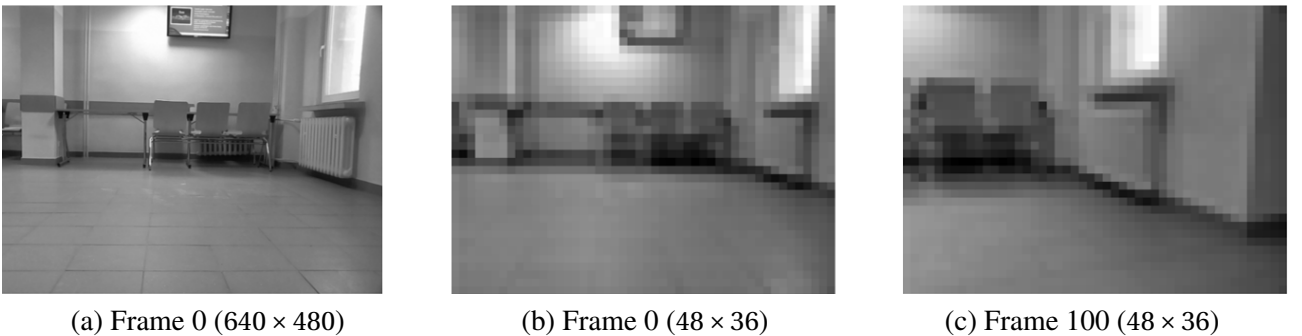


Figure 3: Sample frames from the MDDR sequence illustrating the resolution downsampling and corridor movement. (a) The initial reference frame at full 640×480 pixel resolution. (b) The same initial frame downsampled to 48×36 pixels to reduce computational overhead. (c) Frame 100 downsampled to 48×36 pixels, demonstrating the visual change after robot movement.

During the first 20 frames, the robot remains entirely stationary. Although these initial images appear visually identical to the human eye, yielding a Structural Similarity Index Measure (SSIM) of approximately 0.99, direct pixel-to-pixel comparisons at the full 640×480 resolution reveal that only 3.4% to 22% of the pixels actually match. This underlying variance makes this stationary period highly useful for benchmarking the similarity of the encoded hypervectors. Finally, to mitigate the significant computational overhead that high image resolutions impose on hyperdimensional processing, all images were downsampled from their native 640×480 resolution to 48×36 pixels using the LANCZOS resampling filter from the Python Pillow (PIL) library prior to encoding.

The performance of the encoding techniques (CBFE, RBFE, and BBFE) is heavily influenced by the size of the spatial features extracted by the sliding window. Using the 800-image sequence from the MDDR dataset, we analysed the effects of varying feature dimensions, specifically column length, row length, and block size. For all evaluations discussed in this section, the greyscale flip probability was fixed at 0.01. This value was established to provide an optimal balance: it ensures neighbouring greyscale intensities share enough similarity to mitigate realistic camera noise (such as auto-exposure drift), while preventing the hypervectors from blending into complete uniformity. In these evaluations, each frame is encoded into a composite hypervector and compared against the reference hypervector of the initial frame (Image 0). The sequence consists of a stationary period for the first 20 frames, where high Hamming similarity (> 0.8) is expected despite camera noise, followed by physical robot movement, where similarity should ideally drop to ≈ 0.5 (orthogonality) to indicate that the scene has changed.

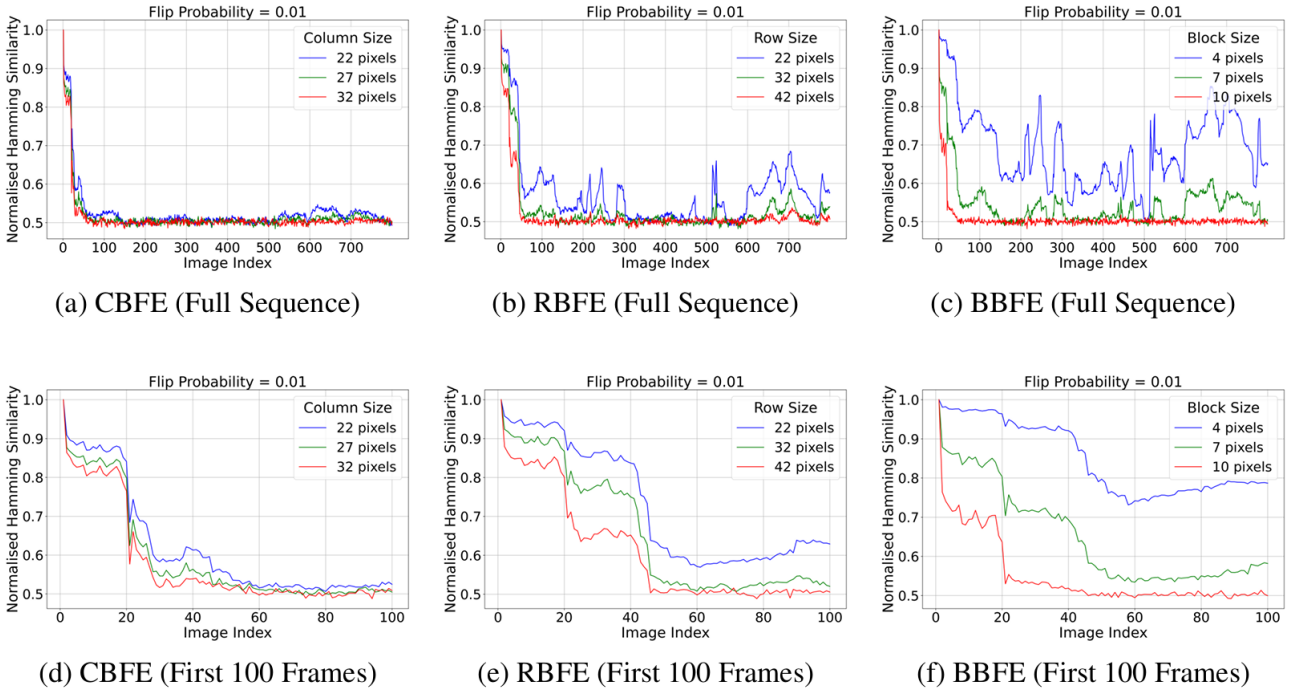


Figure 4: The effect of varying feature dimensions on normalized Hamming similarity. The top row (a, b, c) shows the full 800-image sequence for each encoding method, while the bottom row (d, e, f) provides a truncated view of the first 100 frames to highlight the system’s sensitivity during the transition from a stationary camera to physical movement. All plots maintain a constant greyscale flip probability of 0.01.

Column-Based Feature Encoding (CBFE): Figure 4(a) illustrates the impact of column size. When a shorter feature column of 22 pixels (blue line) is used, the system exhibits spurious recognition later in the sequence, with similarity frequently spiking above 0.5 when it should be orthogonal. Increasing the column size to 32 pixels (red line) successfully suppresses this noise. The truncated plot reveals that for the 32-pixel column, similarity remains high during the stationary phase (frames 0–20) before sharply and consistently

dropping to ≈ 0.5 once movement begins, indicating reliable differentiation of new scenes.

Row Bundled Feature Encoding (RBFE): RBFE demonstrates a higher susceptibility to false similarity than CBFE. As shown in Figure 4(b), horizontal feature extraction struggles to maintain orthogonality with shorter rows; both the 22-pixel and 32-pixel row sizes show significant false positive spikes throughout the 800-frame sequence. To achieve the stable baseline orthogonality (≈ 0.5) seen in the CBFE method, RBFE requires a significantly larger feature window, specifically a row size of 42 pixels (red line).

Block Bundled Feature Encoding (BBFE): The 2D spatial extraction of BBFE presents a distinct sensitivity profile, as seen in Figure 4(c). A small block size, indicated as 4 pixels (blue line), fails entirely to differentiate the shifting corridor, resulting in extreme false-positive recognition (similarity ranging from 0.6 to > 0.8) long after the robot has moved. Conversely, a block size of 10 pixels (red line) perfectly suppresses this late-sequence noise, remaining strictly orthogonal for non-matching images. However, the truncated plot highlights the theoretical trade-off of 2D block encoding: the high structural fidelity of a larger block makes it highly sensitive to the natural camera noise present even in the first 20 stationary frames, causing a much sharper initial drop in similarity compared to the 1D methods.

In the context of Visual Place Recognition (VPR), determining the spatial extent a robot can travel before the current view diverges from the reference image is critical for establishing an efficient keyframe capture strategy. To maintain continuous localisation, a new reference image must be captured just before the normalised Hamming similarity degrades to the orthogonal noise floor of approximately 0.5. Based on the optimal, noise-suppressing feature dimensions identified in Figure 4 (the red plots), the permissible travel distance varies significantly depending on the encoding technique used. As the robot is stationary for the first 20 frames, physical translation only commences at frame 21. Block Bundled Feature Encoding (BBFE) with a 10-pixel block exhibits extreme structural sensitivity; upon movement, its similarity drops to the 0.5 noise floor almost immediately (within 5 to 10 frames of motion). This indicates that while BBFE is highly precise, it requires dense, frequent keyframing. Column-Based Feature Encoding (CBFE) with a 32-pixel column provides a moderate tolerance to movement and maintains a similarity above the noise threshold for approximately 15 to 20 frames of movement before bottoming out near frame 40. Conversely, Row Bundled Feature Encoding (RBFE) with a 42-pixel row demonstrates the widest perceptual tolerance to the forward motion characteristic of this corridor dataset. It sustains a similarity above 0.6 for nearly 25 frames of continuous movement, only decaying to the noise floor around frame 45. Consequently, while BBFE enforces strict spatial bounds ideal for pinpoint accuracy, RBFE allows for a much sparser mapping strategy, significantly reducing the memory footprint required over extended travel trajectories.

Overall, the findings highlight a fundamental trade-off in hyperdimensional feature extraction. While expanding feature dimensions is necessary across all methods to reliably suppress false positives and ensure non-matching images are recognised as orthogonal, it simultaneously amplifies the system’s sensitivity to inherent sensor noise during static scenes. This vulnerability is most pronounced in multi-dimensional block encoding (BBFE), which enforces strict structural fidelity at the cost of rapid similarity degradation. Consequently, the choice of encoding method directly dictates the operational strategy for Visual Place Recognition: highly sensitive methods like BBFE offer pinpoint precision but demand dense, frequent keyframing, whereas one-dimensional approaches like RBFE provide wider perceptual tolerance, enabling a more memory-efficient, sparse mapping strategy.

5 Conclusion

This work introduced Bundled Feature Encoding as a lightweight Hyperdimensional Computing method to improve robustness in visual recognition tasks. By extracting structural information via sliding windows, Column-Based (CBFE) and Row Bundled Feature Encoding (RBFE) successfully provide targeted spatial invariance, overcoming the strict pixel-to-pixel limitations of Whole Picture Pixel Encoding (WPPE). Notably, the one-dimensional extraction of CBFE and RBFE provides greater perceptual tolerance and spatial invariance compared to the rigid, two-dimensional Block Bundled Feature Encoding (BBFE), which suffers from significant computational overhead and extreme sensitivity to movement.

Theoretically and empirically, the orientation of the feature window dictates the system’s targeted invariance. CBFE demonstrates resilience to horizontal perspective shifts, performing well when the robot turns. Conversely, RBFE exhibits superior robustness to vertical camera panning and pitch changes. Crucially, in forward-translating environments, RBFE demonstrates the widest spatial tolerance, sustaining accurate image recognition significantly longer than CBFE or BBFE as the robot advances.

While hyperdimensional encoding remains sensitive to small variations in lighting and camera auto-exposure micro-adjustments, this was successfully mitigated by tuning the greyscale flip probability (e.g., 0.01) to increase intensity similarity, alongside expanding the spatial dimensions of the feature windows. However, our analysis reveals a fundamental trade-off: while larger feature dimensions successfully suppress false-positive recognition between differing scenes, they simultaneously amplify the system’s fragility to static sensor noise. This trade-off directly informs the operational strategy for real-world applications such as Visual Place Recognition (VPR). The strict structural fidelity of BBFE demands dense, frequent keyframing, whereas the wider tolerance of RBFE enables a highly memory-efficient, sparse mapping strategy. Overall, the results demonstrate that one-dimensional feature bundling provides the most balanced, efficient, and reliable approach for lightweight robotic vision systems.

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References

- [Gayler, 1998] Gayler, R. W. (1998). Multiplicative binding, representation operators & analogy. In *Advances in Analogy Research: Integration of Theory and Data from the Cognitive, Computational, and Neural Sciences*, pages 1–4.
- [Kanerva, 1996] Kanerva, P. (1996). Binary spatter-coding of ordered k-tuples. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 1112 LNCS:869–873.
- [Kleyko et al., 2019] Kleyko, D., Gayler, R. W., and Osipov, E. (2019). Learning sensorimotor control with neuromorphic sensors: Toward hyperdimensional active perception. *Science Robotics*, 4:1.
- [Kleyko et al., 2023a] Kleyko, D. V., Rachkovskij, D. A., Osipov, E., and Rahimi, A. (2023a). A survey on hyperdimensional computing aka vector symbolic architectures, part i: Models and data transformations. *ACM Computing Surveys*, 55(6).
- [Kleyko et al., 2023b] Kleyko, D. V., Rachkovskij, D. A., Osipov, E., and Rahimi, A. (2023b). A survey on hyperdimensional computing aka vector symbolic architectures, part ii: Applications, cognitive models, and challenges. *ACM Computing Surveys*, 55(9):175.
- [Mitrokhin et al., 2019] Mitrokhin, A., Sutor, P., Fermüller, C., and Aloimonos, Y. (2019). Learning sensorimotor control with neuromorphic sensors: Toward hyperdimensional active perception. *Science Robotics*.
- [Plate, 1995] Plate, T. A. (1995). Holographic reduced representations. *IEEE Transactions on Neural Networks*, 6:623–641.
- [Schlegel et al., 2022] Schlegel, K., Neubert, P., and Protzel, P. (2022). A comparison of vector symbolic architectures. *Artificial Intelligence Review*, 55:4523–4555.
- [Wozniak et al., 2025] Wozniak, P., Krzeszowski, T., and Kwolek, B. (2025). Multi-domain indoor dataset for visual place recognition and anomaly detection by mobile robots. *Scientific Data* 2025 12:1, 12:817–.